

MATH9102: Fundamentals of Data Analysis

Week 3: Hypothesis Testing, Preparing Data, and Correlation

Today

1. **Hypothesis Testing** — the reasoning every test in this module uses
2. **Assessing Normality** — and why the usual thresholds cannot decide it for you
3. **Preparing Data** — bias, missingness, outliers, transformation
4. **Correlation** — the first real test, and the first real report

The big week

Everything from here on is a variation on what happens today. Week 4 compares two groups, week 5 compares more, week 8 fits a line — all of them with this same procedure around them.

Where We Are

You already have

- A question, a theory, a hypothesis, and named variables (week 1)
- The sampling distribution, the standard error and the CLT (week 2)
- z-scores, and turning a value into a tail probability (week 2)

Today adds

- The formal machinery around that tail probability
- How to decide whether a variable can carry a parametric test
- Your first complete analysis, end to end

You have already done the hard part once

Last week you took a comprehension score, standardised it, and read off how surprising it was. That is a hypothesis test. Today we give the pieces their names.

Hypothesis Testing

The Statistical Model

Every model in this module has one shape.

$$\text{outcome}_i = \text{model} + \text{error}_i$$

For a straight line:

$$y = b_0 + b_1x + \varepsilon$$

- y — the outcome
- x — the predictor
- b_0 — the intercept, y when x is zero
- ε — how far each point sits from the line

What the whole subject is

Choose a model, fit it, and then decide whether the pattern it found is more than noise. Everything else is detail about which model and which test.

Relationship or Difference?

Before you build a predictive model you establish that there is something to predict.

You assert a relationship

- As x rises, y rises or falls consistently
- Test for evidence of association, and judge its strength and direction
- **This week**

You assert a differential effect

- Group A and group B experience the outcome differently
- Test for evidence of a difference, and judge its size
- **Week 4**

Both use the same machinery

Only the statistic changes. The reasoning around it is identical, and it is what the rest of this section is about.

What a Test of Hypotheses Is

The definition

A method that uses **sample data** to decide between two competing claims about a **population**.

The question it answers:

- Is the value I computed a value that would *commonly* occur through natural sampling variation?
- Or is it **surprising** — one that would not easily occur by chance?

The goal, stated plainly

To **rule out chance** as a plausible explanation for what you found. Not to prove your theory — to remove the cheapest alternative to it.

The Two Hypotheses

Hypotheses

H_0 , the null. There is no effect, no difference, no relationship in the population. It always contains an **equality**.

H_1 , the alternative. The competing claim, and what your theory predicts. It always contains an **inequality**.

The procedure

1. *Assume* H_0 holds
2. Collect the data
3. Ask how surprising the data would be if that assumption were true

Why start from the null

It is the claim that specifies a definite value, so it is the only one you can compute a distribution for. “No difference” is a number. “Some difference” is not.

Deriving a Hypothesis Pair

Claim	Hypothesis pair	Tails
Americans work a mean of 34 years before retiring	$H_0: \mu = 34; H_1: \mu \neq 34$	two
The average July high in Dublin has historically been 19°C — perhaps it is higher now	$H_0: \mu \leq 19; H_1: \mu > 19$	one
The mean starting salary for computer science graduates is above £40,000	$H_0: \mu \leq 40000; H_1: \mu > 40000$	one

Watch out

A bare statement of fact is neither hypothesis on its own. It becomes H_0 only once you decide what would count against it — and the wording of the claim is what tells you how many tails you need.

One Tail or Two

A factory makes light bulbs and claims a mean lifetime of 1,000 hours.

Two-tailed

You want to know whether the mean *differs* from 1,000 — longer or shorter, both matter.

$$H_0: \mu = 1000 \quad H_1: \mu \neq 1000$$

One-tailed

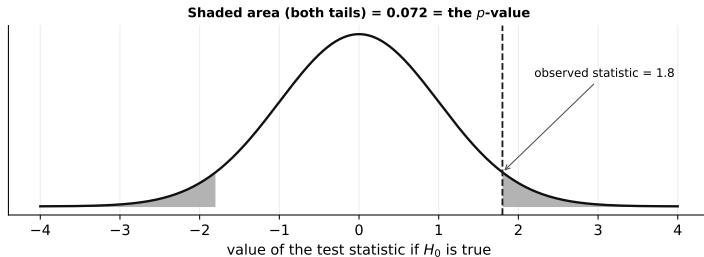
You only care whether they last *longer* than claimed. A shorter lifetime is not a finding you will act on.

$$H_0: \mu \leq 1000 \quad H_1: \mu > 1000$$

Choose before you look

A one-tailed test is easier to pass, and choosing the direction after seeing which way the data went converts that into a free result. The default in this module is **two-tailed**, unless you can justify otherwise in writing.

The p -Value



The definition, and every clause matters

The probability of observing a test statistic **at least as extreme** as the one you got, **assuming the null hypothesis is true**.

It is a probability about *data given a hypothesis*. It is **not** the probability that the null hypothesis is true.

Choosing α

α is the threshold you fix *in advance*: the probability you are willing to accept of rejecting a true null hypothesis.

$\alpha = .05$	a 5% chance of a false positive
$\alpha = .01$	a 1% chance
$\alpha = .001$	a 0.1% chance

Which one is a convention of your field, not a fact.

Reporting convention

Either way, report the actual value to three decimals, without a leading zero: " $p = .032$ ", " $p = .061$ ".

Below .001, write $p < .001$ — more decimals there are spurious precision. "Not significant" on its own says far less than a number.

The confidence interval is the same idea from the other side. With $\alpha = .05$ you build a 95% interval: repeat the study many times and 95% of the intervals you construct would contain the true value. That is a statement about the *procedure*, not about this one interval.

Two Ways to Be Wrong

	H_0 is actually true	H_0 is actually false
You reject H_0	Type I error — a false positive. Probability α	Correct. Probability $1 - \beta$: the power of the test
You do not reject H_0	Correct	Type II error — a false negative. Probability β

Read the first column again: **you choose your own Type I error rate**. That is what α is.

Watch out

And holding everything else constant, lowering α raises β . You cannot reduce both kinds of error by being stricter.

The probability that a test correctly rejects the null when the alternative is true — its ability to detect an effect that is really there.

$$\text{power} = 1 - \beta$$

Power rises with

- The sample size
- The size of the real effect
- A larger α
- Lower measurement error

Why it matters to you

An underpowered study that finds nothing has told you nothing. This is the other half of week 1's “how big should the sample be” — big enough that a real effect of the size you care about would show up.

Two Conclusions You Must Not Draw

A non-significant result

Does *not* mean H_0 is true, and does *not* prove it. It means this sample did not provide sufficient evidence against it.

A significant result

Does *not* mean H_1 is true, and does *not* prove it. It means this sample provided sufficient evidence to reject H_0 in its favour.

And the caveat that follows

Is $p = .049$ really a different finding from $p = .051$? **No.** The threshold makes decisions repeatable; it is not a line in nature. Report the actual value and an effect size, and let the reader see how close it was.

Assessing Normality

Parametric or Non-Parametric?

	Parametric	Non-parametric
Assumes a distribution for the population	yes — usually normal	no
Suits	interval or ratio data	ordinal data, skewed data, small samples
Power when its assumptions hold	higher	lower — may miss real effects

So the decision procedure is

Look at the variable, decide whether the normal model is **close enough for what you are about to do**, and say what you decided from.

“Non-parametric” does not mean assumption-free. It means no assumption about the *shape*; the design assumptions are unchanged.

How to Inspect a Continuous Variable

In this order, every time.

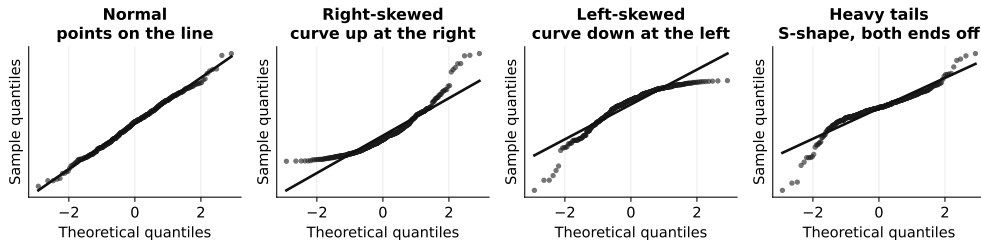
1. A **histogram** with a normal curve over it
2. A **Q-Q plot**
3. Central tendency and dispersion
4. Skew and kurtosis, standardised
5. The percentage of standardised scores outside the usual bounds

The order is the argument

Steps 1 and 2 make the decision. Steps 3 to 5 *describe* what you decided about, and go into the write-up so a reader can disagree with you.

Then report the statistics appropriate to what you concluded: mean and SD if you are treating it as normal, median and IQR if not.

Reading a Q-Q Plot



It plots your data's quantiles against the quantiles a normal distribution would have produced. Agreement means the points lie on the line.

Four pictures

Learn these and you can read any Q-Q plot you meet. Look for *trends* in the departure, not for individual stray points.

The Standardised Skew Ratio

```
# The standardised skew ratio: skew divided by its standard error.  
x = survey.tpcoiss.dropna()  
n = len(x)  
  
skew = stats.skew(x, bias=False)  
se_skew = np.sqrt(6 * n * (n - 1) / ((n - 2) * (n + 1) * (n + 3)))  
  
print(f"skew = {skew:.3f}, standardised = {skew / se_skew:.2f}")  
  
# The SE depends only on n, so the ratio grows with n for a fixed shape.
```

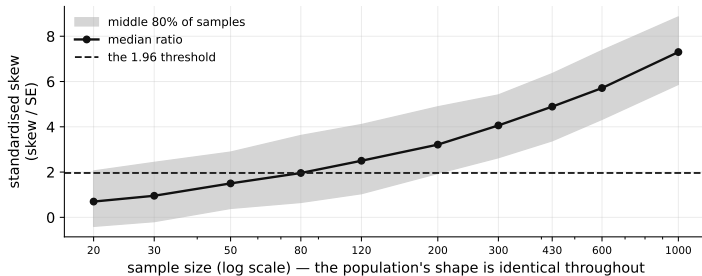
Compare the ratio against ± 1.96 , and the same for kurtosis. Cited to Field, Miles and Field (2012) and to Curran, West and Finch (1996).

Both are reasonable *descriptions*. Neither is a decision rule, and the next slide is why.

A second heuristic

Count the standardised scores beyond the bounds, against what normality predicts: about 5% past ± 1.96 , 1% past ± 2.58 , 0.1% past ± 3.29 . One case past ± 3.29 earns a look.

Why the Threshold Cannot Decide It



One population, one fixed shape, throughout. The standard error of skew depends only on n , so the ratio grows with the sample while the distribution does not change at all.

At this module's sample sizes

A survey of four hundred will fail the test for a shape that a survey of fifty would pass.

Normality Tests Have the Same Problem

The usual suspects

- **Shapiro–Wilk** — small samples
- **Kolmogorov–Smirnov** — larger ones
- **Anderson–Darling** — weights the tails more
- **D’Agostino–Pearson** — combines skew and kurtosis

Watch out

All of them test the null hypothesis of *exact* normality — which no real variable satisfies. At a large n they will reject it for departures far too small to affect anything, and at a small n they lack the power to detect departures that matter.

The question they cannot answer

Not “is this normal?” — nothing is — but “is it close enough for what I am about to do?” That is a judgement about consequences, and no test knows what you are about to do.

Our Convention

```
# Visual first. The panel is a histogram with a normal curve and a Q-Q plot.  
m9.normality_panel(survey.tpcoiss.dropna(), label="Total perceived control")  
  
# Then the numbers, as descriptions of shape rather than as a verdict.  
m9.report_normality(survey, "tpcoiss", "Total perceived control")
```

What the generated narrative does

It reports the shape statistics **with the sample size beside them**, and states the sensitivity explicitly. The reader gets your evidence and your reasoning, and can disagree with the second without recomputing the first.

Preparing Data

It Is All About the Bias

A sample is biased when it differs *systematically* from the population — some group over-represented, some under.

- **Selection** — using only certain cases or groups
- **Sampling** — non-random selection; a fact of life with convenience samples
- **Non-response** — those who did not answer differ from those who did
- **Time interval** — a collection window that is not representative
- **Confirmation** — choosing data that supports what you already believe
- **Omitted variable** — leaving a relevant concept out

None of these is fixed by a larger sample

More data collected the same biased way is more biased data. Recognise it, reduce it where you can, and report what remains.

Two things are certain in quantitative research: measurement error and missing data.

What it costs

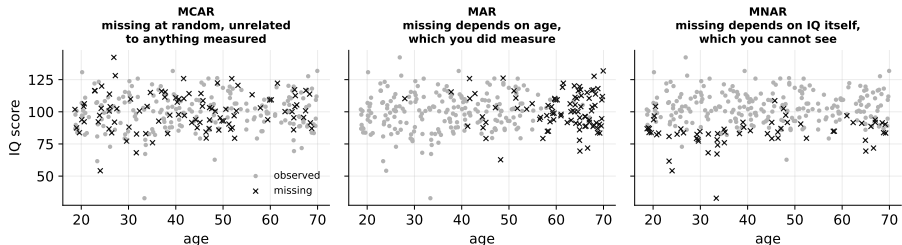
- Reduced statistical power
- Reduced representativeness
- **Bias in the estimates**, in a direction you cannot see
- Some tests become inappropriate or awkward

The key fact

The **mechanism** and the **pattern** matter more than the amount. A dataset missing 2% for a systematic reason is more dangerous than one missing 10% at random.

A useful benchmark (Tabachnick and Fidell): under about 5%, missing randomly, in a large dataset, and almost any treatment gives similar results.

Three Mechanisms

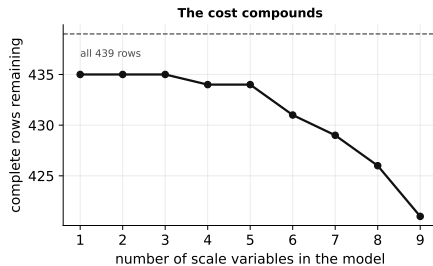
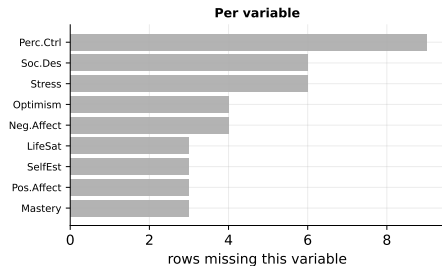


MCAR can be ignored. **MAR** may be modelled, because the thing driving it is something you measured. **MNAR** is not ignorable.

You cannot test for MNAR

The evidence you would need is exactly the data you do not have. Argue it from what you know about the collection.

Diagnosing It



Look at the amount *per variable*, the combinations across variables, and whether the rate differs by group. Little's test can support a claim of MCAR — but only after you have ruled out MNAR yourself.

The right-hand panel

Each variable you add to the model costs more rows. This is the number that decides the analysis.

Counting the Cost

```
# What listwise deletion costs, for the variables the model actually uses.
scales = ["toptim", "tmast", "tposaff", "tnegaff", "tlifesat",
          "tpstress", "tslfest", "tmalow", "tpcoiss"]

everything = m9.missingness(survey, scales)
print(everything.attrs["summary"])

two_only = m9.missingness(survey, ["tpcoiss", "tpstress"])
print(two_only.attrs["summary"])
```

Select the variables, then count

Run complete-case deletion across all 134 columns of this dataset and you throw away a large share of the sample — most of it to variables that appear in none of your models. Count the cost for the variables you named.

What To Do About It

Deletion

- **Listwise** — drop the case if any needed variable is missing. Simple; costs power, and biases the result if the data are not MCAR
- **Pairwise** — use each case wherever it has the values needed. Keeps more data; the sample size then varies between analyses, which complicates the standard errors

Imputation

- **Mean substitution** — cheap, and it shrinks the variance and weakens correlations *by construction*
- **Regression imputation** — better, still too confident
- **Maximum likelihood** and **multiple imputation** — unbiased estimates and honest standard errors. What to reach for on a real project

Whatever you choose

Report it. The treatment is part of the method, and a reader cannot judge your result without it.

Weighting

A correction applied when you know your sample's composition differs from the population's.

For non-response

- Compare the proportion surveyed in each group against the population proportion
- The weight is expected divided by actual
- Multiply that group's responses by it in every statistic

For design effects

- Cluster sampling — survey 100 schools rather than chase 2,000 scattered pupils
- Pupils within a school resemble each other, so the sample carries less information than its size suggests
- Ignore it and you understate the variability: **increased risk of a Type I error**

The correct treatment for clustering is multilevel modelling, beyond this module. Check whether your dataset ships a weighting variable, and find out which of the two it corrects for.

Outliers, Again

A second screen: standardised scores

- Sample of 80 or fewer: beyond ± 2.5
- Larger than 80: beyond ± 3.29
- Interval and ratio variables only — and ordinal ones you are treating as metric

Univariate and multivariate

- **Univariate** — unusual on one variable
- **Multivariate** — unremarkable on each variable separately, but a rare *combination*
- Someone 150 cm tall is ordinary.
Someone 150 cm tall weighing 140 kg is not

Reasons, and what each implies

A data-entry error gets corrected. A missing-value code read as data gets recoded. A case that is not really a member of your population gets excluded, and you say so. A genuine extreme member **stays** — run the analysis with and without it and report if the conclusion changes.

Transformation

```
# If a variable is badly skewed you can transform it - but check the result,  
# and remember the transformed variable is what you will be interpreting.  
skewed = survey.tnegaff.dropna()  
logged = np.log(skewed + 1)  
  
for name, x in [("raw", skewed), ("log(x + 1)", logged)]:  
    print(f"{name}: skew = {stats.skew(x, bias=False):+.3f}")  
  
# A transformation that does not help is a transformation you do not apply.
```

Log — pulls in a right tail; first choice for positive skew. **Square root** — gentler; suits counts and values near zero. **Reciprocal** — strong; for large outliers. **Reverse then transform** — for negative skew.

If transformation does not help, use a test that does not need normality.

Watch out

Check that it worked. You are now interpreting the transformed variable — a coefficient on $\log(\text{income})$ is not one on income. And some values are not permissible: $\log 0$, division by zero, the root of a negative.

Correlation

From Covariance to Correlation

Variance measures how one variable deviates from its own mean. **Covariance** does it for two at once.

1. For each case, its deviation from the mean on x
2. And its deviation from the mean on y
3. Multiply the two
4. Average the cross-products

Positive: the two rise together. Negative: one rises as the other falls.

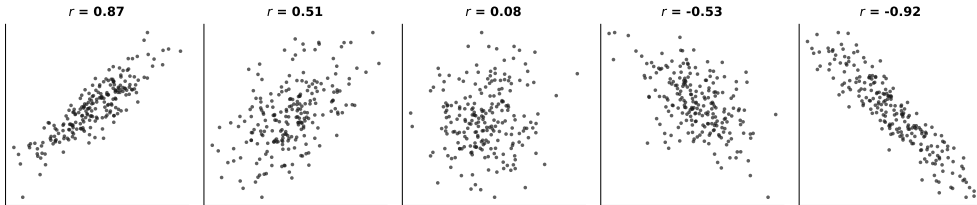
But the size means nothing

Covariance depends on the units. The same relationship measured in miles gives one number and in kilometres another.

Correlation is standardised covariance

Divide by both standard deviations and it is forced between -1 and $+1$: -1 a perfect negative straight line, $+1$ a perfect positive one, 0 no *linear* relationship.

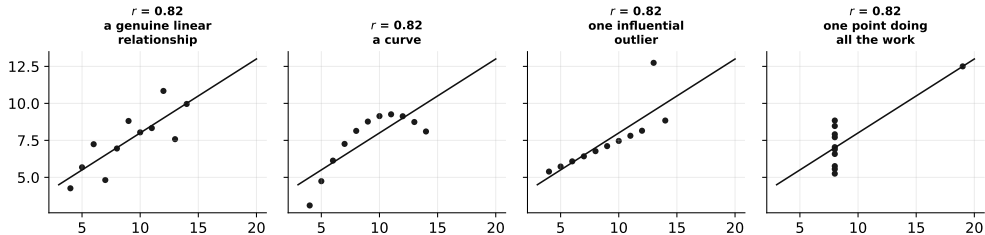
What Each Value of r Looks Like



Read three things off a scatterplot: the **direction** (does it slope up or down), the **strength** (how tightly do the points hug the line), and the **form** (is it a line at all).

An outlier far from the trend weakens r even when everything else is tightly linear — which is one reason to look at the picture before trusting the number.

Four Datasets, One Correlation



Anscombe's quartet. Identical means, identical variances, identical correlations, identical fitted lines.

Only the first is a correlation worth reporting

And no summary statistic could have told you that.

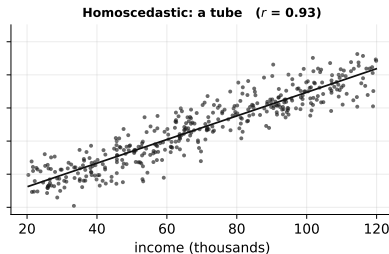
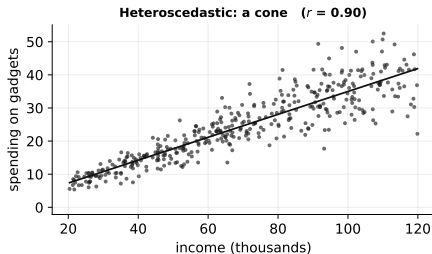
Assumptions of Pearson's r

1. **Two interval or ratio variables.** One may be binary, with roughly equal group sizes
2. **Related pairs** — each case supplies a value on both
3. **Independent observations** — no case influences another
4. **Normality** — inspect each variable
5. **Linearity** — inspect the scatterplot. Pearson measures straight-line association and nothing else
6. **Homoscedasticity** — constant spread around the line

On independence

Study the television habits of children and take several children from each family, and the observations are not independent — siblings resemble each other. No amount of care with the other five assumptions repairs that.

Homoscedasticity



Income and spending on gadgets. People on low incomes all spend little; people on high incomes vary enormously. The spread grows with x .

Watch out

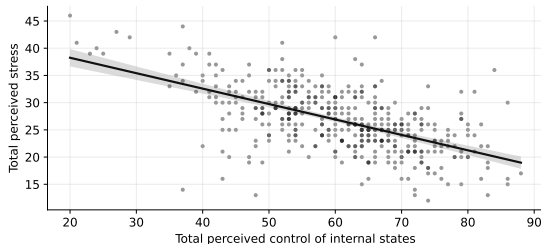
Same r in both panels. In the left one a prediction for a high earner is far less reliable than for a low earner — and one coefficient cannot tell you that.

A Worked Correlation

Hypotheses

H₀: There is no relationship between a respondent's perceived ability to regulate their internal states and their level of perceived stress.

H₁: There is a relationship. Two-tailed, $\alpha = .05$.



Linear, roughly constant spread, no isolated influential point, both variables scale totals. **Pearson.**

Running It

```
# Plot first, then test. Anscombe's quartet is the reason this order matters.
m9.scatter_with_fit(survey, "tpcoiss", "tpstress",
                    xlabel="Total perceived control",
                    ylabel="Total perceived stress")

pairs = survey[["tpcoiss", "tpstress"]].dropna()
result = stats.pearsonr(pairs.tpcoiss, pairs.tpstress)

m9.report_correlation(survey, "tpcoiss", "tpstress", result,
                     x_label="perceived control of internal states",
                     y_label="perceived stress")
```

Why a *t*-statistic appears

The test of “is this correlation different from zero?” is a *t*-test on $n - 2$ degrees of freedom.

Watch out

Degrees of freedom are not the sample size. They differ by two here. Report n as n and df as df — quoting one as the other misstates the study.

Unlike most test statistics, r is directly interpretable. Ignore the sign for magnitude.

$\pm.1$	small / weak
$\pm.3$	medium / moderate
$\pm.5$	large / strong

Cohen's conventions. Cite a source respected in your field when you use them.

r^2 : the coefficient of determination

Square r and you get the **proportion of variance the two variables share**.

A correlation of .2 shares 4% of its variance. That is the number that stops a "significant" weak correlation being oversold.

Statistical Against Practical Significance

Small samples

- A moderate correlation may fail to reach significance
- The relationship is there; you cannot demonstrate it

Large samples

- A negligible correlation will reach significance easily
- It is still negligible

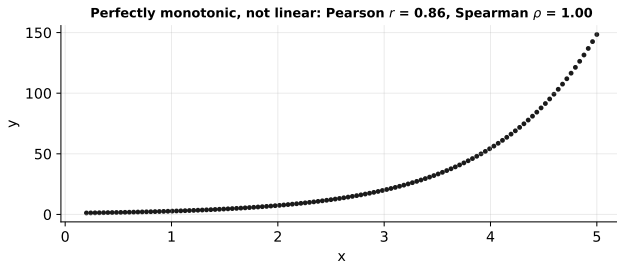
Report all three

The coefficient, the shared variance, and the p -value. The first two say how big; the third only says whether you can claim it is there.

And compare against your field

A finding that explains a few per cent of the variance may be poor or impressive depending on what others have found. Say which.

When Pearson Is Not Appropriate



Spearman's ρ correlates the *ranks*, so it measures **monotonic** association: does one rise as the other rises, by any shape of curve?

Kendall's τ counts concordant and discordant pairs. More robust; more often the two agree than not.

Use either when Pearson's assumptions fail or the data are not scale. Spearman is more widely reported; check which your discipline prefers.

Ties Are Not Missing Data

```
# Spearman ranks the data, so it measures monotonic association rather than
# linear association. Kendall counts concordant and discordant pairs.
rho = stats.spearmanr(pairs.tpcoiss, pairs.tpstress)
tau = stats.kendalltau(pairs.tpcoiss, pairs.tpstress)

print(f"Spearman rho = {rho.statistic:.3f}, p = {rho.pvalue:.3g}")
print(f"Kendall tau = {tau.statistic:.3f}, p = {tau.pvalue:.3g}")

# Tied values are ordinary in questionnaire data. Both functions correct for
# them automatically. Ties are not an error, and they are not missing data.
print(f"tied values in tpcoiss: {pairs.tpcoiss.duplicated().sum()}")
```

Watch out

Questionnaire totals produce many tied values, and you will read that this breaks Spearman. It does not: ties are handled by a correction applied automatically, and in older R code they produce a **warning** about exact p -values, not an error. **They are not missing data.** And run **one** coefficient — reporting the friendliest of several is fishing.

Every correlation report needs all of this:

- Which coefficient, and on what variables
- The inspection that justified the choice
- The coefficient, to two or three decimals
- n , and separately df if you quote it
- p , by the reporting convention
- The shared variance, if it is relevant
- An interpretation in ordinary language

Naming the statistic

Pearson: r . Spearman: ρ , r_s , or “Spearman’s rho”. Kendall: τ , or τ_b .

Say which you used. Two of them are routinely written r .

Correlation Is Not Causation

r measures association. It is symmetric, and it knows nothing about time, mechanism or direction.

What it cannot tell you

- Which variable causes which
- Whether a third variable causes both
- Whether the relationship is causal at all

Week 1's random walks

Two independent series correlated strongly because both trended over time. Nothing in the arithmetic knew they were unrelated, and nothing in the arithmetic ever will.

Watch out

Watch your own wording. “An increase in x *leads to* an increase in y ” is a causal claim written by accident. “Respondents scoring higher on x also tended to score higher on y ” is what you are entitled to.

Close

What you should take from this week

You SHOULD be able to

- State a hypothesis pair, choose a tail direction in advance, and fix α before looking
- Define a p -value precisely, and say what it is not
- Distinguish Type I from Type II error, and say what power is
- Assess normality visually first, and explain why the standardised ratios and normality tests cannot decide it at this sample size
- Name the missing-data mechanisms, argue which one you are facing, and choose a treatment
- Screen for outliers two ways, and decide what a flagged value means
- Check linearity and homoscedasticity before computing a correlation
- Run, interpret and report a Pearson or Spearman correlation with its effect size and shared variance
- Say what a correlation does not license you to claim

That's all for this week